

12

Understanding the Function of Behavior

INTRODUCTION

When Tier I and II interventions are unsuccessful at remediating problematic behaviors, more intensive, individualized interventions are required. This chapter describes techniques used to hypothesize reasons for serious behavior problems.

Behavioral experts spend years learning and mastering functional behavioral assessment (FBA) and functional behavioral analysis (FBA). It is not within the scope of this chapter to provide comprehensive instruction on FBA or FBA. Rather, the intent is to provide a broad overview of FBA to an educational community that has not benefited from intensive training on behavioral theory, research, assessment, and modification.

We begin with the premise that FBA is a means of gathering information related to the environment, antecedents, and consequences that cause or permit a problem behavior (Drasgow, Yell, Bradley, & Shiner, 1999; Gresham, Watson, & Skinner, 2001).

CONCEPTS TO BE INTRODUCED

- Functional behavioral assessment (FBA)
- Indirect and direct assessment methods
- Positive reinforcement
- Negative reinforcement
- Functional behavioral analysis (FBA)

LINKS TO PREVIOUS DISCUSSIONS

Previous chapters delineated means for identifying students needing individualized interventions. This chapter details procedural necessities for FBA implementation. Conducting meaningful FBAs is time

intensive and reserved for students for whom Tier II interventions were not effective, or whose behaviors are of a severity that they require immediate intensive attention. The FBA determines the function, or purpose, of a behavior and the interventions most likely to elicit pro-social behaviors from the target student.

FORMS TO BE INTRODUCED

[Form 12.1](#): Target Peer Comparison Form for a Functional Behavioral Assessment (FBA)

[Form 12.2](#): Conditional Probability Form

[Form 12.3](#): Functional Behavior Assessment Form—Teacher Interview

[Form 12.4](#): Antecedent–Behavior–Consequence (Narrative Version)

[Form 12.5](#): Antecedent–Behavior–Consequence (Checklist Version)

MATERIALS NEEDED

Cumulative file Previous records of interventions Office discipline referrals Academic data

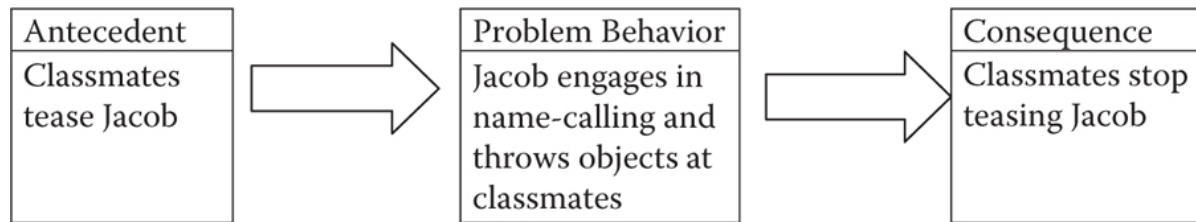
PEOPLE NEEDED

- Student's teacher
- Student
- Student's parent or guardian
- Other informants who are knowledgeable about the student (e.g., bus driver, cafeteria worker, and people from the community)

INFORMATION NEEDED: UNDERSTANDING FBA

FBA information analyzes the reasons behind a student's specific problem behaviors. Within this analysis, pattern identification suggests the function of the problem behavior. To identify these patterns, FBAs generally approach problem behavior by analyzing a three-term relationship: antecedent, behavior, and consequence (A-B-C).

Example of an A-B-C relationship:



Outcome: Jacob is more likely to engage in name-calling and throwing objects when classmates tease him.

Outcome: Jacob is more likely to engage in name-calling and throwing objects when classmates tease him.

Figure 12.1 This chart demonstrates the A (antecedent), B (behavior), C (consequences) relationship.

Antecedents are precipitating events (e.g., person, time of day, place, or demand or request) occurring prior to the behavior of interest. A “triggering” antecedent typically occurs immediately before the behavior.

Target behaviors are the observable and measurable inappropriate behaviors of interest that need to be suppressed, and the appropriate behaviors that should increase.

Consequences are events (e.g., social, tangible, activity, and sensory stimulation) following the student’s target behavior that increase or decrease the likelihood of the behavior occurring. (See [Figure 12.1](#).)

After examining the behavior chain, the problem-solving team concluded that an appropriate intervention would entail teaching Jacob to ignore, walk away, and/or tell an adult when classmates tease him. In other words, the desirable behaviors will produce the same result as the problem behavior but without the additional negative consequences (e.g., being sent to the office, or injuring others). Once Jacob learned to ignore classmates and get adult assistance, the classmates stopped teasing him. To ensure that Jacob was motivated to acquire the skill and fluently assimilate it as a means of getting his classmates to stop teasing him, the teacher rewarded Jacob when he exhibited the desirable behavior and admonished the offending students, thus further reducing the likelihood of teasing, name calling, and throwing projectiles within the classroom.

(See [Figure 12.2](#).)

Antecedents

As stated previously, antecedents are events or conditions that occur before the student behaves inappropriately, and they often serve as triggers. A particular set of questions, answered through interviews and observations, helps to identify triggering antecedents. The questions are as follows:

- When does the problem behavior usually occur ?
- Where does the problem behavior usually occur?
- Who is present when the problem behavior occurs?
- What activities or events precede the occurrence of the problem behavior?
- What do other people say or do immediately before the problem occurs?
- What do other people say or do immediately after the problem occurs?
- Does the student engage in any other behaviors before the problem behavior?
- When, where, with whom, and in what circumstances is the problem behavior least likely to occur?

Behavior

As described earlier, behavior is the student's actions—what the student says or does. Defining the behavior in observable and measurable terms is the most critical component of the FBA process. If the target behaviors lack clear and accurate definitions, the entire FBA process can be invalidated. For instance, an ambiguous definition may result in overreporting a single behavior that actually represents multiple behaviors. In addition, antecedents and consequences of the poorly defined behavior become less meaningful. The following are characteristics of clear behavioral definitions:

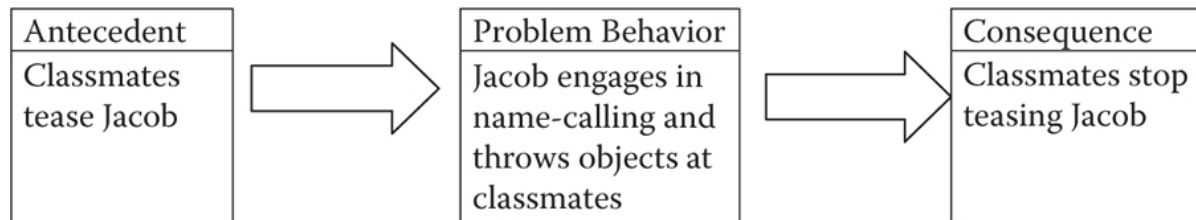
Active verbs: Behavioral definitions describe what a student is *saying* or *doing*.

Unambiguous: There should not be multiple interpretations of a behavioral definition. For instance, a behavior described as *unsportsmanlike conduct* in baseball could mean yelling obscenities, throwing the bat at someone, refusing to shake hands, or kicking the dirt.

Observable: Internal states such as anger, sadness, or apathy cannot be directly observed and should be excluded from behavioral definitions. Similarly, perceived intentions should also be excluded from behavior definitions. *Describe one or two behaviors:* Target the most problematic behaviors. It becomes logistically more difficult to track increasing numbers of problematic behaviors. Once the first targeted problem behavior improves, the team can redirect its focus to challenging secondary problem behavior.

Example of an A-B-C relationship following successful intervention:

Example of an A-B-C relationship:



Outcome: Jacob is more likely to engage in name-calling and throwing objects when classmates tease him.

Outcome: Jacob is more likely to walk away, ignore, and get adult assistance when classmates tease him.

Figure 12.2 This chart provides an example of an A-B-C relationship following a successful intervention.

Examples of well-defined behaviors:

- Sean's *tantrums* are recorded as occurring whenever he cries and sobs, lies on the floor and kicks the floor or wall, or pounds toys or other objects on the floor.
- Michael *hits* with his fists (versus "Michael is aggressive").
- Keisha is *studying* whenever she is observed underlining sentences in the text, completing math or physics workbook exercises, copying notes from class, or outlining chapters from the text.
- Lisa *takes* money from her classmates (versus "Lisa is delinquent").
- Alison is *sharing* whenever she passes or hands material to another classmate, when she exchanges materials with another classmate, or when she uses the same material at the same time with another classmate (e.g., she and another classmate color on the same piece of paper).
- Byron *arrives 10 minutes late* (versus "Byron is irresponsible").
- Danielle is *out of her seat 75% of the time* (versus "Danielle is hyperactive").

Consequences

A consequence describes what happens as a result of the student's behavior. It helps to maintain the behavior. To determine a behavior's consequence, ask the following questions:

- What happens after the problem behavior occurs?
- What do the school staff members do when the problem behavior occurs?
- What do other students do when the problem behavior occurs?
- What changes in the environment after the problem behavior occurs?
- What does the student receive after the problem behavior?

- What does the student get out of or avoid after the problem behavior?

The consequences of behavior generally fall into one of four categories.

Positive Reinforcement

Positive reinforcement increases the probability of a behavior and occurs when a target behavior is followed by a positive stimulus. For instance, a student turning in a lost wallet may receive a reward or recognition for his honesty. A second example is receiving a cola after pressing a button on a soda machine. Positive reinforcement involves the presentation of a stimulus that increases the likelihood of a behavior.

Negative Reinforcement

Negative reinforcement also increases the probability of a behavior and occurs when a target behavior is followed by removal, escape, or avoidance of an aversive stimulus. For instance, a student with poor reading skills may act out before his turn to read orally and be sent to the office without reading aloud to the class. A second example is changing TV channels from an undesirable channel. Negative reinforcement involves the removal of a stimulus that increases the likelihood of a behavior.

Positive Punishment

Positive punishment decreases the probability of a behavior and occurs when a target behavior is followed by presentation of an aversive stimulus. For instance, a student is caught talking during a test and receives a grade of F as a consequence. A second example is getting bit when attempting to pet an angry dog. Positive punishment involves the presentation of a stimulus that reduces a behavior.

Negative Punishment

Negative punishment decreases the probability of a behavior and occurs when a target behavior is followed by removal of a positive stimulus. For instance, a student may lose recess for not completing an in-class assignment. A second example is taking away a teen's cell phone for exceeding his or her plan's monthly minutes. Negative punishment involves the removal of a stimulus, which decreases a behavior.

Identifying Problem Behaviors, Antecedents, and Consequences

Multiple steps are necessary to complete an FBA. Although these are presented in a linear order, one should understand that the steps may not fall in lockstep with one another. An indirect assessment may lead to a direct assessment, but an evaluator may need to conduct more interviews after watching the child. Throughout the entire process, remember to evaluate the environment—what is happening around the child that is both eliciting problem behaviors and preventing desirable behaviors.

Indirect Assessment

Indirect assessment refers to file reviews and interviews that provide background information about a child's behavior. The advantage of indirect assessment is that a great deal of information can be collected quickly. However, the information may not contain adequate specificity and is subject to interviewee bias. We recommend the following steps:

- Review the child's file. [Form 12.1](#) may be used when reviewing comprehensive files. Pay close attention to ODRs, previous special education forms including previous FBAs and behavior intervention plans (BIPs), medical history, grade history, and academic or other assessment information.
- A member of the problem-solving team completes the Functional Behavior Assessment Form—Teacher (FBAF) with the child's teacher. The form is located in Appendix A (see [Form 12.3](#)). Middle school problem-solving teams complete this form with the teachers identifying the student's most challenging behaviors.
- A member of the problem-solving team interviews the student to get his or her perspective on the problematic behavior, and then identifies reinforcers when conducting a preference assessment.
- The parent interview records the parental perspective of the problem behavior and any relevant background information.
- Interview other relevant personnel (e.g., a bus driver or outside service provider).
- Review the student's cumulative folder (e.g., grades, test scores, health record, and discipline record).

- Review Tier II intervention data in addition to any other interventions that may have been implemented informally.

The indirect assessment component of an FBA involves gathering, in a written format, information from the student and people familiar with the student. A structured interview is the best method to obtain information from the student, parent(s), teacher(s), outside service providers, bus driver, and others who are able to provide meaningful information about the student and behaviors of concern.

The Functional Behavior Assessment Form—Teacher (FBAF) is an instrument used to interview the teacher. It requires a behavioral intervention expert to meet with the teacher and ask questions regarding the behaviors of concern, possible antecedents, and possible consequences. The student interview will involve asking the student a series of questions to ascertain why he or she is engaging in the behavior(s) of concern. In addition, the interviewer can conduct a preference assessment to learn what is rewarding for the student so that effective reinforcers can be identified for the intervention plan. Jim Wright delineates a method involving a reward deck (Wright, n.d.). A *reward deck* is a tool that teachers can use in collaboration with the problem-solving team representative and parent to quickly select and regularly update student reward menus. This strategy involves five steps:

1. The teacher reviews a list of reward choices typically available in school settings. (Teachers can use the comprehensive sampling of possible school rewards that appears in Appendix D.) From this larger list, the teacher selects only those rewards that she or he approves of using, believes would be acceptable to other members of the school community (e.g., administration and parents), and finds feasible and affordable. The list should include rewards that allow a student access to preferred tangibles, access to student or teacher attention, or avoidance of difficult assignments and difficult relationships. (*Note:* Sometimes a student may be reluctant to volunteer items that he or she enjoys or wants. It is also reasonable to observe the student's behavior when there are no expectations, and watch the types of behaviors he or she demonstrates. Is the student trying to get adult or peer attention? Is he or she playing alone? Are there some activities he or she seems to enjoy doing? What are the high-frequency behaviors during free time?)
2. From the initial list, he or she “screens” the reward choices for any reinforcers that seem inappropriate for the particular student. For example, the teacher may screen out the reward “pizza party” because the magnitude of the reinforcer is too great (and too expensive) when paired with a minor difficulty involving homework completion.
3. The teacher writes out the acceptable reward choices on index cards to create a master “reward deck.”

4. The teacher sits with the student and presents each of the reward choices remaining in the reward deck. For each reward option, the student indicates whether he or she (a) likes the reward a lot, (b) likes the reward a little, or (c) doesn't care for the reward. The teacher sorts the reward options into three piles that match these rating categories. The teacher can then assemble that student's reward menu using the student's top choices ("like a lot"). If the teacher needs additional choices to fill out the rest of the menu, he or she can pull items from the student's "like a little" category as well. Periodically (e.g., once a month), the teacher should meet with the student and repeat the above procedure to make sure the rewards are still reinforcing to the student. At this point, some rewards may need to be removed and replaced with others.
5. Finally, the problem-solving team representative reviews the archival data in the student's cumulative folder (e.g., grades, test scores, health record, and discipline record)

Direct Assessment

The direct assessment component of an FBA utilizes school staff personnel with experience in conducting structured observations (e.g., a behavior specialist or school psychologist). The problem-solving team representative records each instance of problematic behavior along with its antecedents and consequences. The data from the indirect assessment component help to determine which behaviors to target along with the possible antecedents and consequences. The next step involves selection of an observational method. Recommended methods may include conditional probability assessments and A-B-C records.

The conditional probability method involves tracking when behaviors occur and what happens before and after the behavior. Next, the observer calculates what percentage of the time a certain event precedes or follows the problematic behavior. If an event or condition consistently precedes a behavior, then it is likely that the event or condition is sustaining the problematic behavior. A procedure for using the conditional probability method is located in Appendix A (see [Form 12.3](#)).

The A-B-C method has three possible formats: descriptive, checklist, or interval record. The descriptive format involves recording observations as a narrative under the columns for antecedent, behavior, and consequence and time it occurred.

The checklist format has columns for antecedents, behaviors, and consequences based on the indirect assessment data. If the observer puts many possible antecedents, behaviors, and consequences into the form, he or she will do less writing than with the narrative format. The possible antecedents, behaviors, and consequences are already listed, and the observer simply records the identity of who is being

observed, notes the time the observation starts, and places a mark under which antecedent, behavior, and/or consequence occurs. (See [Figure 12.4](#).)

Date/Time	ANTECEDENT What happened just <i>before</i> the behavior?	BEHAVIOR What behavior happened? Be specific.	CONSEQUENCE What happened just <i>after</i> the behavior?
7/7/09 10:00 a.m.	Teacher asked the class to take out their reading books to prepare to take turns reading aloud.	Robert pushed back his chair, stood up, and kicked the chair over.	The teacher told Robert to go to the principal's office. The other students gasped.

Figure 12.3 Example of A-B-C narrative format (see [Form 12.4](#)).

Person	Time	Antecedents		Behavior			Consequences		
		Reading Lesson	Group Activity	Screams	Hits another person	Kicks his desk	Peers Give Attention	Sent out of the Classroom	Teacher Gives Attention
J.W.	10:00 a.m.		X	X					
J.W.	10:05 a.m.	X				X		X	X

Figure 12.4 Example of A-B-C checklist format (see [Form 12.5](#)).

The interval format involves dividing an observation period into brief time intervals and recording whether a particular behavior, antecedent, or consequence occurred during that interval. It is a more detailed form than the above described checklist in that it yields information about the frequency, duration, and sequence of behaviors. (See [Figure 12.5](#).) The reading lesson was the sole antecedent during the observation. It occurred 60% of the time (i.e., 6/10, because the observation ended when the student left the room). The student left his seat, talked to a peer, and kicked his desk during the reading lesson. The teacher redirected the student twice—once after he got out of his seat and once after he talked to another peer. After the second redirection, the student kicked his desk and consequently was sent out of the room by the teacher. The student engaged in problematic behaviors in 30% of the observation periods.

- A problem-solving team representative identifies in measurable and observable terms the problematic behaviors and their possible antecedents and consequences based on the indirect assessment data.
- A problem-solving team representative identifies, based on indirect assessment data, when and where to observe the problematic behavior.
- A problem-solving team representative chooses an observational method that will yield information about the behavior and its antecedents and consequences (conditional probability observation form or program; A-B-C records).
- A problem-solving team representative evaluates observational data and develops hypotheses about the function of the behavior.

	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5	Total
Possible Antecedents																
Reading lesson				X	X	X	X	X	X							6
Lunch time																
P.E. class																
Peer says something																
Request is refused																
Behaviors																
Kicks desk									X							1
Gets out of seat					X											1
Talks to another peer							X									1
Hits another peer																
Possible Consequences																
Teacher redirects the student						X		X								2
Peer speaks to the student																
Student is sent out of the room										X						1

Figure 12.5 Example of A-B-C interval format (see Form 12.6).

Hypothesis Testing

Once the behavioral analyst determines the hypothesized function of a behavior, a hypothesis statement should be written. This statement includes the following pieces of information: the child, the problematic behavior, a statement of the settings where the behavior occurs, and a function of the behavior. Some hypothesis statements include the following:

- When Jordan is given directions to start his work by his teacher, he begins to yell back at his teacher to escape engaging in the assignment.
- In all class settings, Michelle calls other students names because she gains peer attention.
- When given work that is difficult for him, Jonathan throws chairs in the class because he is able to escape the task.
- As lunchtime approaches and Charla's Ritalin begins to lose its effect on her concentration, Charla begins to shout out answers to questions in order to gain teacher attention.

These hypothesis statements give clear indications about the antecedent, the behavior, and the consequence of that behavior.

Functional Analysis (FBAn)

The FBAn involves the deliberate manipulation of variables to determine the effect those variables have on a student's behavior. The functional analysis helps to confirm a hypothesis, or helps to choose between two or more possible functions of behavior. In schools, this process, normally called *brief functional analysis*, can greatly reduce the amount of time spent on interventions that are ineffective because they address the wrong functional needs for a child. Sterling-Turner, Robinson, and Wilczynski (2001) described a functional analysis of a child who was throwing spitballs. Following the functional assessment, the team hypothesized that the child's behavior was motivated by attention. However, they were unsure whether he sought teacher attention or peer attention. They then ran two separate experiments. In the first experiment, the teacher was asked to ignore the child's spitball throwing. The behavior persisted despite the removal of teacher attention. Next, Sterling-Turner et al. (2001) recruited

a cohort of students and asked them to ignore the child's spitball-throwing behavior. When the students were able to successfully ignore the behavior of the student demonstrating the problematic behavior, that child's spitball behavior quickly extinguished. Thus, the researchers determined that the behavior was motivated by peer attention, and created interventions designed to alleviate that concern.

Setting Analysis

A type of experimental analysis that is closely related to functional analysis includes a setting analysis. Behavior specialists utilizing experimental analysis alter environmental triggers to determine which setting events are most likely serve as a catalyst for a behavior. For example, members of a behavior team may guess that a teacher's directives are triggering a child's verbally explosive behaviors, but they are unsure which types of directives provoke this behavior. They ask the teacher to give the child directives in close proximity to the child but find that the child's behavior persists. Next, the team has the teacher give nonverbal directives whereby both she and the student are trained to give signals for "Get back to work,"

"Turn and face forward," "Thank you," and "You're doing a good job." They found that nonverbal directives substantially reduced the occurrence of the child's disruptive behaviors. Either form of experimental analysis can provide valuable information about which interventions may be effective, which setting events trigger problematic behaviors, or what events can make a desirable behavior more likely.

CONCLUSION

Functional assessments yield correlational rather than causal data, but allow for greater levels of objective data to be collected and utilized. Correlational data allow one to develop a strong hypothesis about the antecedents and consequences of behavior. When a behavior consistently precedes or follows a particular event, condition, or behavior, it becomes clear where to intervene. If a student's calling out in class always results in laughter from his peers, then the intervention should focus on extinguishing the reinforcing laughter while teaching the student alternative and appropriate ways to gain peer attention. Members of the problem-solving team should review all FBA data and generate a hypothesis regarding the function of the behavior, which typically falls into one of two broad categories: obtain something, or avoidance or escape.

After identifying the function of a behavior, the team creates a BIP designed to decrease the problematic behavior and increase the student's use of an acceptable alternative behavior that serves the same function.